

DDA5002 Optimization

Lecture 13 Projected Gradient Method

Other Algorithms

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Final Exam

- **Date and Time:** December 15 (Monday), 10am–12pm
- **Location:** Liwen Hall
- Closed-book and closed-notes
- No cheat sheet
- Five questions covering:
 - Branch-and-Bound Algorithm 1
 - Optimality Conditions, KKT, and Convexity 1
 - Algorithms (bisection, golden section, gradient descent, Newton's method, projected gradient method) 2
 - One question from material covered before the midterm 1
- Sample final is posted and will be discussed during next week's tutorial.
- Homework 6 (last homework) is due next Friday (December 12).
- The final review session will be held during next Wednesday's class.

Outline

- 1 Gradient Descent Review
- 2 Newton's Method
- 3 Projected Gradient Method
- 4 Why the Projected Gradient Method Works
- 5 Stochastic Gradient Descent
- 6 Subgradient
- 7 Subgradient Method

- 1 **Gradient Descent Review**
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Gradient Descent Algorithm

Start with any point $\mathbf{x}^0$. Set $k = 0$ and stopping criterion $\epsilon > 0$

- 1 Check $\|\nabla f(\mathbf{x}^k)\|$. If $\|\nabla f(\mathbf{x}^k)\| \leq \epsilon$, stop and output $\mathbf{x}^k$. Otherwise, continue to Step 2
- 2 Let $\mathbf{d}^k = -\nabla f(\mathbf{x}^k)$
- 3 Use either exact line search or backtracking line search to find α_k
- 4 Let $\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha_k \mathbf{d}^k$, let $k = k + 1$. Go back to step 1.

Step Size

Descent direction: function value decreases with small stepsize
 $1, 0.1, 0.01$ $\{d: \nabla f(x)^T d < 0\}$

- Constant step size: $\alpha_k = \bar{\alpha}$ for all k .
- Exact line search: choose α_k such that

$$\alpha_k = \arg \min_{\alpha} \underline{f(\mathbf{x}^k + \alpha \mathbf{d}^k)} := g(\alpha) \quad \text{one dimension}$$

- Backtracking / Armijo line search:

- 1 Let $\sigma, \gamma \in (0, 1)$ be given. Start with $\alpha = 1$.
- 2 If $f(\mathbf{x}^k + \alpha \mathbf{d}^k) \leq f(\mathbf{x}^k) + \gamma \alpha \cdot \nabla f(\mathbf{x}^k)^T \mathbf{d}^k$, choose $\alpha_k = \alpha$.
Otherwise, set $\alpha = \sigma \alpha$ and repeat this step.

→ Armijo condition

shrink α by σ
 $f(\mathbf{x}^k + \alpha \mathbf{d}^k) < f(\mathbf{x}^k)$ $1 \rightarrow \sigma \rightarrow \sigma^2 \rightarrow \sigma^3 \rightarrow \dots$

Lipschitz Continuity and Smoothness

- A function $f : \Omega \rightarrow \mathbb{R}$ is **Lipschitz continuous** over Ω with constant $L > 0$ if:

$$|f(x) - f(y)| \leq L\|x - y\|, \quad \forall x, y \in \Omega.$$

- A differentiable function f is **Lipschitz smooth** over Ω if its gradient ∇f is Lipschitz continuous with constant $L > 0$:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y \in \Omega.$$

- If $f : \Omega \rightarrow \mathbb{R}$ is twice continuously differentiable and Ω is convex, then f is Lipschitz smooth iff $\lambda_{\max}(\nabla^2 f(x)) \leq L$

$$\|\nabla^2 f(x)\| \leq L \quad \text{for all } x \in \Omega.$$

- Descent Lemma:** If $f : \Omega \rightarrow \mathbb{R}$ is continuously differentiable and L -Lipschitz smooth, where Ω is convex, then for any $x, y \in \Omega$:

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2.$$

Convexity of Functions

Definition: Convex Function

A function $f : \Omega \rightarrow \mathbb{R}$ is **convex** if, for all $x, y \in \Omega$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$



Definition: Strictly Convex Function

A function $f : \Omega \rightarrow \mathbb{R}$ is **strictly convex** if, for all $x \neq y$ in Ω and $\lambda \in (0, 1)$:

$$f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y).$$

Definition: Strongly Convex Function

A function $f : \Omega \rightarrow \mathbb{R}$ is **strongly convex** with parameter $m > 0$ if, for all $x, y \in \Omega$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) - \underbrace{\frac{m}{2} \lambda(1 - \lambda) \|x - y\|^2}_{\text{red wavy underline}}.$$

First-Order Characterizations of Convexity

Assume $f : \Omega \rightarrow \mathbb{R}$ is differentiable and Ω is convex.

Convex Function

f is **convex** if and only if, for all $x, y \in \Omega$:

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x).$$



Strictly Convex Function

f is **strictly convex** if and only if, for all $x, y \in \Omega$ and $x \neq y$:

$$f(y) > f(x) + \nabla f(x)^\top (y - x).$$

Strongly Convex Function

f is **strongly convex** with parameter $m > 0$ if and only if, for all $x, y \in \Omega$:

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|^2.$$

Second-Order (Hessian) Characterizations

Assume $f : \Omega \rightarrow \mathbb{R}$ is twice continuously differentiable and Ω is convex.

Convex Function

f is **convex** if and only if, for all $x \in \Omega$:

$$\nabla^2 f(x) \succeq 0$$

$\nabla^2 f(x)$ is positive semidefinite. $\lambda_{\min}(\nabla^2 f(x)) \geq 0$

Strictly Convex Function

f is **strictly convex** if, for all $x \in \Omega$:

$$\nabla^2 f(x) \succ 0$$

$\nabla^2 f(x)$ is positive definite. $\lambda_{\min}(\nabla^2 f(x)) > 0$

Strongly Convex Function

f is **strongly convex** with parameter $m > 0$ if and only if, for all $x \in \Omega$:

$$\nabla^2 f(x) \succeq mI$$

$\lambda_{\min}(\nabla^2 f(x)) \geq m.$

Gradient Descent Convergence Rates

Consider unconstrained problem $\min_{x \in \mathbb{R}^n} f(x)$.

- **Sublinear convergence:** If f is convex and ∇f is Lipschitz continuous, GD with constant step size has sublinear convergence (in function value) to the global minimizer.
- **Linear convergence:** If f is strongly convex and ∇f is Lipschitz continuous, GD with constant step size / exact / Armijo line search will have linear convergence to the unique global minimizer.
- **Global convergence:** No matter where GD starts, it will always converge to a local minimizer (when $f(\cdot)$ is convex, it converges to a global minimizer).

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Quadratic Function Intuition

- Let x^k be the current iterate and consider a second-order Taylor approximation of $f(x)$ at x^k , i.e.

$$\overset{\text{>}}{g}(x) = f(x^k) + \nabla f(x^k)^\top (x - x^k) + \frac{1}{2}(x - x^k)^\top \nabla^2 f(x^k)(x - x^k).$$

- Choose the next iterate as the solution that minimizes the approximate function $g(x)$.
- Setting $\nabla g(x) = 0$, we get the linear system

Newton System

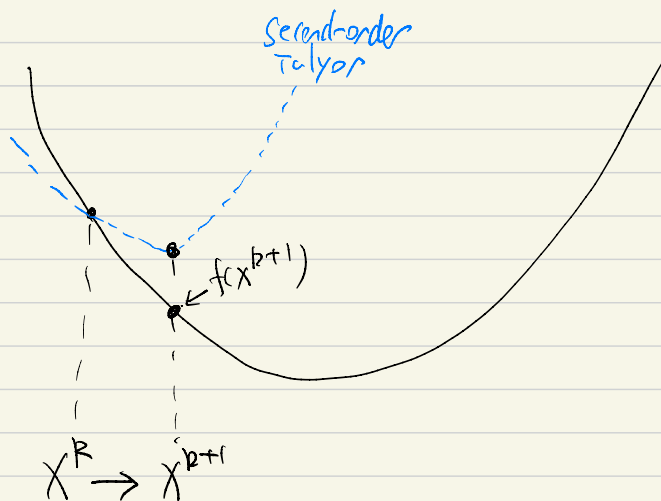
Solving next $\rightarrow$ $\nabla f(x^k) + \nabla^2 f(x^k)(x - x^k) = 0.$

- If the Hessian is non-singular, a solution to the above system is

$$\rightarrow x^{k+1} = x^k - [\nabla^2 f(x_k)]^{-1} \nabla f(x^k).$$

- Hence, if the original objective function is a quadratic function, then Newton's method converges in one step.

Taylor Approx



Complete Procedure of Newton's Method

Start from a point $\mathbf{x}^0$, set tolerance $\epsilon > 0$ and $k = 0$.

- 1 Compute the Newton's step:

$$\rightarrow \mathbf{d}^k = -(\nabla^2 f(\mathbf{x}^k))^{-1} \nabla f(\mathbf{x}^k)$$

If $\|\nabla f(\mathbf{x}^k)\| < \epsilon$, then output $\mathbf{x}^k$, otherwise continue;

- 2 Choose step size α_k by backtracking line search or set by 1.
- 3 $\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha_k \mathbf{d}^k$, $k = k + 1$, go back to step 1.

 backtracking
line search

Linear Convergence vs Quadratic Convergence

Remember the gradient descent method can have linear convergence rate (f is strongly convex, ∇f is Lipschitz continuous):

$$|x^{k+1} - x^*| \leq \eta \underbrace{|x^k - x^*|}_{\text{distance}}.$$

Now, Newton's method has quadratic convergence rate (if initial point is sufficiently close to x^* , f is strong convex, $\nabla^2 f$ is Lipschitz continuous):

$$|x^{k+1} - x^*| \leq C \underbrace{|x^k - x^*|^2}_{\text{distance}^2}.$$

Newton's Method is Fast in Iteration Count, But ...

①
It requires computing the second-order derivative (Hessian) in each iteration.

- ② It might be computationally expensive to do so, especially if the second-order derivative doesn't have a closed-form (which is the case for many useful applications).
- ③ *more memory* It also requires a lot of space to store the Hessian matrix (If it is an n -dimensional problem, then we need n^2 space, comparing to n space required for gradient descent method).
- ④ It requires much more matrix computation in each iteration.
- ⑤ If the Hessian is singular (or close to singular) at some iteration, we cannot proceed. *non-invertible*
- ⑥ Newton's method may diverge, even for convex functions.

⑦

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Constrained Optimization

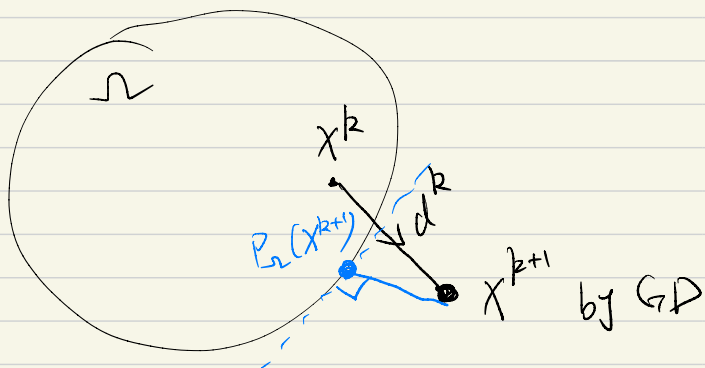
We consider the following constrained optimization problem:

$$\begin{aligned} & \rightarrow \min_{\mathbf{x}} f(\mathbf{x}) \\ & \text{subject to } \mathbf{x} \in \Omega. \end{aligned}$$

where $\Omega \subset \mathbb{R}^n$ is a convex and closed set.

In the methods for unconstrained problems, the main idea is:

- At each $\mathbf{x}^k$, compute a descent direction $\mathbf{d}^k$.
- Then find an appropriate step size α_k and update $\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha_k \mathbf{d}^k$.
- Both the gradient and Newton's method are based on this basic idea.
- For constrained problem, $\mathbf{x}^{k+1}$ can become infeasible. One solution is to project the point back to Ω .



Euclidean Projection / Orthogonal Projection

Definition: Euclidean / Orthogonal Projection

x is given

Let $\Omega \subset \mathbb{R}^n$ be a nonempty, closed, convex set. The (Euclidean / orthogonal) projection of $\mathbf{x}$ onto Ω is defined as the unique minimizer $\mathbf{y}^*$ of the constrained optimization problem:

$$\min_{\mathbf{y}} \quad \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|^2 \quad \text{s.t.} \quad \mathbf{y} \in \Omega$$

and we write $\mathbf{y}^* = \mathcal{P}_{\Omega}(\mathbf{x})$.

Observations:

- Interpretation: Projection is to find the closest point $\mathbf{y}^*$ in Ω to $\mathbf{x}$.
- Hence, the projection $\mathbf{y}^* = \mathcal{P}_{\Omega}(\mathbf{y})$ is the point in Ω that has the minimum distance to $\mathbf{x}$.

$$\mathcal{P}_{\Omega}(\mathbf{x}) = \mathbf{x} \quad \text{if } \mathbf{x} \in \Omega$$

Example I: Box Constraints

Suppose that Ω is given by box constraints:

$$\Omega = \{\mathbf{x} \in \mathbb{R}^n : x_i \in [a_i, b_i], \forall i\} = [\mathbf{a}, \mathbf{b}],$$

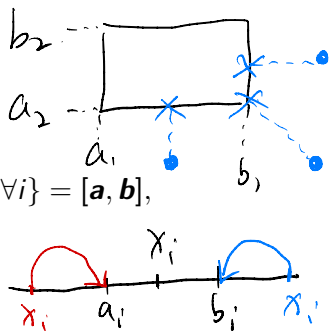
where $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, $\mathbf{a} \leq \mathbf{b}$, are given.

Euclidean Projection:

- The projection onto Ω can be computed as follows:

$$[\mathcal{P}_\Omega(\mathbf{x})]_i = \mathcal{P}_{[a_i, b_i]}(x_i) = \max\{\min\{x_i, b_i\}, a_i\}, \quad \forall i.$$

$$\mathcal{P}_{[a_i, b_i]}(x_i) = \begin{cases} a_i & \text{if } x_i < a_i \\ x_i & \text{if } a_i \leq x_i \leq b_i \\ b_i & \text{if } x_i > b_i \end{cases}$$



Example II: Linear Constraints

We now consider the simple case where Ω consists of linear equality constraints:

$$\Omega = \{\mathbf{x} : \mathbf{Ax} = \mathbf{b}\},$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$ are given.

Euclidean Projection:

- Suppose that $\mathbf{A}$ has full row rank ($m \leq n$), then it holds that:

$$\mathcal{P}_{\Omega}(\mathbf{x}) = \mathbf{x} - \mathbf{A}^{\top}(\mathbf{AA}^{\top})^{-1}(\mathbf{Ax} - \mathbf{b}).$$

- This yields a special case when $\Omega = \{\mathbf{x} : \underbrace{\mathbf{a}^{\top} \mathbf{x} = \mathbf{b}}\}$, i.e., projection onto a hyperplane. We have

$$\mathcal{P}_{\Omega}(\mathbf{x}) = \mathbf{x} - \frac{\mathbf{a}^{\top} \mathbf{x} - \mathbf{b}}{\|\mathbf{a}\|^2} \mathbf{a}.$$

*take-home
exercise*

$$\Omega = \{x: Ax=b\}, \quad x \text{ is given}$$

$$\left. \begin{array}{l} \min. \quad \frac{1}{2} \|y-x\|^2 \\ \text{s.t.} \quad Ay=b \end{array} \right\} \Rightarrow \text{convex OPT}$$

$$\text{Lagrangian: } L(y, \mu) = \frac{1}{2} \|y-x\|^2 + \mu^T (Ay-b)$$

KKT:

$$\text{main: } \nabla_y L(y, \mu) = y - x + A^T \mu = 0$$

$$\text{Complementary slackness: } \mu^T (Ay-b) = 0$$

$$\text{Primal feasible, } Ay=b$$

$$\text{dual feasible: any } \mu$$

$$A(y-x + A^T \mu) = A(0)$$

$$Ay - Ax + AA^T \mu = 0$$

↓

$$b - Ax + AA^T \mu = 0$$

$$\mu = (AA^T)^{-1} (Ax-b)$$

$$\begin{aligned} y &= x - A^T \mu = x - A^T (AA^T)^{-1} (Ax-b) \\ &= P_{\Omega}(x) \end{aligned}$$

Example III: Ball Constraints

Suppose that Ω is a Euclidean ball with radius $r > 0$ and center $\mathbf{m} \in \mathbb{R}^n$, i.e.:

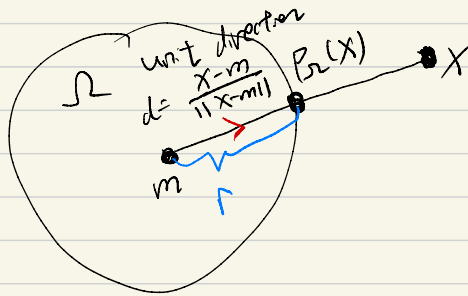
$$\Omega = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{m}\| \leq r\}.$$



Euclidean Projection:

- The projection onto Ω can be computed as follows:

$$\mathcal{P}_{\Omega}(\mathbf{x}) = \begin{cases} \mathbf{x} & \text{if } \|\mathbf{x} - \mathbf{m}\| \leq r, \\ \mathbf{m} + \frac{r}{\|\mathbf{x} - \mathbf{m}\|}(\mathbf{x} - \mathbf{m}) & \text{if } \|\mathbf{x} - \mathbf{m}\| > r. \end{cases}$$



$$P_{\Omega}(x) = \begin{cases} x & \text{if } x \in \Omega \\ m + r \cdot \frac{x-m}{\|x-m\|} & \text{if } x \notin \Omega \end{cases}$$

$$\begin{array}{ll} \min. & \|y-x\|^2 \\ \text{s.t.} & \|y-m\| \leq r \end{array} \rightarrow \begin{array}{l} \text{solve by KKT} \\ \text{exercise} \end{array}$$

$\Leftrightarrow y \in \Omega$

Projected Gradient Method: Basic Idea

In the gradient descent method, we perform a gradient step of the form:

$$\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k - \lambda_k \nabla f(\mathbf{x}^k),$$

where $\lambda_k > 0$ is a step size. *for GD*

Motivation & Strategy:

- Setting $\mathbf{x}^{k+1} = \mathbf{x}^k - \lambda_k \nabla f(\mathbf{x}^k)$ might likely generate infeasible iterates: $\mathbf{x}^{k+1} \notin \Omega$.
- Idea: We project the step $\mathbf{x}^k - \lambda_k \nabla f(\mathbf{x}^k)$ back onto Ω :

$$\mathbf{x}^{k+1} = \mathcal{P}_\Omega(\mathbf{x}^k - \lambda_k \nabla f(\mathbf{x}^k)).$$

Questions:

- How to choose $\lambda_k > 0$? If λ_k is determined by line search, every adjustment of λ_k requires re-evaluating the projection $\mathcal{P}_\Omega(\mathbf{x}^k - \lambda_k \nabla f(\mathbf{x}^k))$. This can be expensive.
- Can we guarantee descent?

Important Observation

Observation:

$$\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k + \alpha_k \mathbf{d}^k$$

$$\mathbf{x}^{k+1} = \mathcal{P}_\Omega(\mathbf{x}^k - \lambda_k \nabla f(\mathbf{x}^k)) = \mathbf{x}^k + \underbrace{\left[\mathcal{P}_\Omega(\mathbf{x}^k - \lambda_k \nabla f(\mathbf{x}^k)) - \mathbf{x}^k \right]}_{\mathbf{d}^k}.$$

- This is close to our usual update form: $\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha_k \mathbf{d}^k$.
- Setting $\mathbf{d}^k = \mathcal{P}_\Omega(\mathbf{x}^k - \lambda_k \nabla f(\mathbf{x}^k)) - \mathbf{x}^k$, we can consider:

$$\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha_k \mathbf{d}^k = (1 - \alpha_k) \mathbf{x}^k + \alpha_k \mathcal{P}_\Omega(\mathbf{x}^k - \lambda_k \nabla f(\mathbf{x}^k)).$$

If $\alpha_k \in [0, 1]$, then the convexity of Ω implies that $\mathbf{x}^{k+1}$ will be feasible if $\mathbf{x}^k \in \Omega$.
because Ω is convex set

Why does / can this work? Is $\mathbf{d}^k$ a descent direction?

The Projected Gradient Method

Projected Gradient Method

min. $f(x)$
s.t. $x \in \Omega \rightarrow$ convex set

- 1 Initialization: Choose an initial point $\mathbf{x}^0 \in \Omega$.
- 2 For $k = 0, 1, \dots$:
 - 1 Select $\lambda_k > 0$ and compute $\nabla f(\mathbf{x}^k)$ and the search direction

stepsize for GD $\rightarrow \mathbf{d}^k = \mathcal{P}_\Omega(\mathbf{x}^k - \lambda_k \nabla f(\mathbf{x}^k)) - \mathbf{x}^k$.

- 2 Choose a step size $\alpha_k > 0$. $\rightarrow$ stepsize for Projected gradient method
- 3 Set $\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha_k \mathbf{d}^k$.
- 4 If $\|\mathbf{d}^{k+1}\| \leq \epsilon$ then STOP and $\mathbf{x}^{k+1}$ is the output.

why? $\rightarrow$ when $\mathbf{d}^k$ is small, $\mathbf{x}^k \approx \mathcal{P}_\Omega(\mathbf{x}^k - \lambda_k \nabla f(\mathbf{x}^k))$

- In terms of convergence theorem, under appropriate conditions on the step sizes α_k and λ_k , projected gradient method has similar convergence guarantees to those of GD.

$\downarrow$
FONC

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Optimization Problems with Convex Constraints

Recall the constrained optimization problem:

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{s.t.} \quad \mathbf{x} \in \Omega, \rightarrow \text{Convex set} \quad (1)$$

where $\Omega \subset \mathbb{R}^n$ is a convex and closed set.

- We can derive the following optimality condition:

Theorem: FONC for Problems with Convex Constraints

Let f be continuously differentiable on an open set that contains the convex and closed set $\Omega \subset \mathbb{R}^n$. Let $\mathbf{x}^* \in \Omega$ be a local minimizer of (1), then:

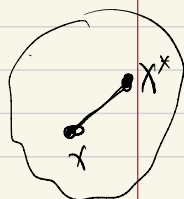
$$\rightarrow \nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) \geq 0, \quad \forall \mathbf{x} \in \Omega. \leftarrow \quad (2)$$

- A point satisfying (2) is called a stationary point of problem (1).

critical point

Proof: Let $x^* \in \Omega$ be a local optimal
 $\exists \varepsilon > 0$ s.t.

$$f(x^*) \leq f(x), \quad \forall x \in B_\varepsilon(x^*)$$



Let $x \in \Omega$ and $\lambda \in [0, 1]$

Define $x(\lambda) = \lambda x + (1-\lambda)x^* \in \Omega$ because Ω is convex set

Rewrite $x(\lambda) = x^* + \lambda(x - x^*)$

$$\text{If } x(\lambda) \in B_\varepsilon(x^*)$$

$$\|x(\lambda) - x^*\| \leq \varepsilon$$

$$\Leftrightarrow \|\lambda x + (1-\lambda)x^* - x^*\| \leq \varepsilon$$

$$\Leftrightarrow \|\lambda(x - x^*)\| \leq \varepsilon$$

$$\Leftrightarrow \lambda \|x - x^*\| \leq \varepsilon$$

$$\Leftrightarrow \lambda \leq \frac{\varepsilon}{\|x - x^*\|}$$

When $\lambda \in [0, \frac{\varepsilon}{\|x - x^*\|}]$, $x(\lambda) \in B_\varepsilon(x^*)$

$$f(x^*) \leq f(x(\lambda))$$

$$\Leftrightarrow \frac{f(x(\lambda)) - f(x^*)}{\lambda} \geq 0 \quad \text{if } \lambda \neq 0$$

$$\Leftrightarrow \lim_{\lambda \rightarrow 0} \frac{f(x^* + \lambda(x - x^*)) - f(x^*)}{\lambda} \geq 0$$

$$\Leftrightarrow \nabla f(x^*)^\top (x - x^*) \geq 0$$

Projection Theorem

Projection Theorem

Let Ω be a nonempty, closed, and convex set. Then:

- 1 • A point $\mathbf{y}^*$ is the projection of $\mathbf{x}$ onto Ω , i.e., $\mathbf{y}^* = \mathcal{P}_\Omega(\mathbf{x})$, if and only if

$$\rightarrow (\mathbf{y}^* - \mathbf{x})^\top (\mathbf{y} - \mathbf{y}^*) \geq 0, \quad \forall \mathbf{y} \in \Omega.$$

- 2 • The mapping $\mathcal{P}_\Omega : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is Lipschitz continuous with constant $L = 1$, i.e.,

$$\|\mathcal{P}_\Omega(\mathbf{x}) - \mathcal{P}_\Omega(\mathbf{y})\| \leq 1 \|\mathbf{x} - \mathbf{y}\|, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

- 3 • (FONC for Problem with Convex Constraints) For any $\lambda > 0$, the vector $\mathbf{x}^*$ is a stationary point of (1) if and only if

$$\nabla f(\mathbf{x}^*) (\mathbf{x} - \mathbf{x}^*) \geq 0 \Rightarrow \mathbf{x}^* - \mathcal{P}_\Omega(\mathbf{x}^* - \lambda \nabla f(\mathbf{x}^*)) = 0.$$

Proof: 1 $y^* = P_\Omega(x)$

m.h. $\frac{1}{2} \|y - x\|^2 = f(y)$

s.t. $y \in \Omega \rightarrow \text{convex}$

Plug in FONC in (2)

$\nabla f(y^*) (y - y^*) \geq 0, \forall y \in \Omega$

$\Rightarrow (y^* - x)^T (y - y^*) \geq 0, \forall y \in \Omega$

$y^* = P_\Omega(x)$

(*)

2. From (*),

$y^* = P_\Omega(x)$

$(P_\Omega(x) - x)^T (z - P_\Omega(x)) \geq 0, \forall z \in \Omega$

$y^* = P_\Omega(y)$

$(P_\Omega(y) - y)^T (z - P_\Omega(y)) \geq 0, \forall z \in \Omega$

Pick

$z = P_\Omega(x)$

Pick

$z = P_\Omega(x)$

$(P_\Omega(x) - x)^T (P_\Omega(y) - P_\Omega(x)) \geq 0$
 $(P_\Omega(y) - y)^T (P_\Omega(x) - P_\Omega(y)) \geq 0$

$\Rightarrow (P_\Omega(x) - P_\Omega(y))^T (P_\Omega(y) - y + x - P_\Omega(x)) \geq 0$

$$\Rightarrow (P_{\Omega}(x) - P_{\Omega}(y))^T (P_{\Omega}(y) - P_{\Omega}(x) + x - y) \geq 0$$

$$\Rightarrow -\|P_{\Omega}(x) - P_{\Omega}(y)\|^2 + (P_{\Omega}(x) - P_{\Omega}(y))^T (x - y) \geq 0$$

$$\Rightarrow \|P_{\Omega}(x) - P_{\Omega}(y)\|^2 \leq (P_{\Omega}(x) - P_{\Omega}(y))^T (x - y)$$

Cauchy

Schwarz
Inequality

$$a^T y \leq \|a\| \|y\|$$

$$\Rightarrow \|P_{\Omega}(x) - P_{\Omega}(y)\|^2 \leq \|P_{\Omega}(x) - P_{\Omega}(y)\| \|x - y\|$$

$$\Rightarrow \|P_{\Omega}(x) - P_{\Omega}(y)\| \leq \|x - y\|$$

So $P_{\Omega}(x)$ is a Lipschitz continuous function.

$$3. \quad x^* = P_{\Omega}(x^* - \lambda \nabla f(x^*))$$

from 1

$$\Leftrightarrow (x^* - (x^* - \lambda \nabla f(x^*)))^T (x - x^*) \geq 0, \forall x \in \Omega$$

Proj

inside
Proj

any pt
in Ω

Proj

$$\Leftrightarrow \lambda \nabla f(x^*)^T (x - x^*) \geq 0, \forall x \in \Omega$$

Since $\lambda > 0$

$$\Leftrightarrow \nabla f(x^*)^T (x - x^*) \geq 0, \forall x \in \Omega$$

↓

FONC

Descent Direction

Lemma: Descent Direction

Let $\mathbf{x} \in \Omega$ and $\lambda > 0$ be given. If $\mathbf{x}$ is not a stationary point of (1), then the direction

$$\mathbf{d} := \mathcal{P}_{\Omega}(\mathbf{x} - \lambda \nabla f(\mathbf{x})) - \mathbf{x}$$

is a descent direction and it holds that

$$\Rightarrow \nabla f(\mathbf{x})^{\top} \mathbf{d} \leq -\frac{1}{\lambda} \|\mathbf{d}\|^2 < 0.$$

$\mathbf{d}$ is called the generalized gradient.

$$\text{Proof: } \nabla f(x)^T d$$

$$= \nabla f(x)^T (P_R(x - \lambda \nabla f(x)) - x)$$

$$= \nabla f(x)^T (P_R(x - \lambda \nabla f(x)) - P_R(x)) \quad \text{||} \rightarrow \text{sgn}$$

$$= -\frac{1}{\lambda} (-\lambda \nabla f(x))^T (P_R(x - \lambda \nabla f(x)) - P_R(x))$$

$$= -\frac{1}{\lambda} (x - \lambda \nabla f(x) - x)^T (P_R(x - \lambda \nabla f(x)) - P_R(x))$$

by
⊗ ⊗
above

$$\leq -\frac{1}{\lambda} \|P_R(x - \lambda \nabla f(x)) - P_R(x)\|^2$$

$$= -\frac{1}{\lambda} \|P_R(x - \lambda \nabla f(x)) - x\|^2$$

$$= -\frac{1}{\lambda} \|d\|^2 < 0$$

End of materials for Final!

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- 1 Gradient Descent Review
- 2 Newton's Method
- 3 Projected Gradient Method
- 4 Why the Projected Gradient Method Works
- 5 Stochastic Gradient Descent**
- 6 Subgradient
- 7 Subgradient Method

Example: solving multiple linear regression

$$\min. \sum_{i=1}^n (Y_i - \beta_1 X_{i1} - \beta_2 X_{i2} - \dots - \beta_{p+1} X_{i,p+1})^2$$

Quadratic function $\min_{\beta} \|y - X\beta\|_2^2$, $X \in \mathbb{R}^{n \times (p+1)}$ $\rightarrow$ convex

$\downarrow$
 $f(\beta) = \|y - X\beta\|_2^2 = (Y - X\beta)^2$

Gradient $\nabla f(\beta) = -2X^T(Y - X\beta) = 2X^T(X\beta - Y)$

Exact solution $\rightarrow \nabla f(\beta) = 0 \Rightarrow \hat{\beta} = (X^T X)^{-1} X^T Y$

Gradient descent

Quiz: Newton's method

The objective function is

$$\beta^{k+1} \leftarrow \beta^k - 2\partial_{\beta} \underbrace{X^T(X\beta^k - Y)}$$

$\rightarrow f(\beta) = \|y - X\beta\|_2^2 = \sum_{i=1}^n (y_i - \beta^T X_i)^2$

which is separable. Most loss functions in machine learning problems are separable.

Newton's Method

$$x^{k+1} \leftarrow x^k - [\nabla^2 f(x^k)]^{-1} \nabla f(x^k)$$

$$x^{k+1} \leftarrow (X^T X)^{-1} X^T y$$

Stochastic gradient descent (GSD)

- ▶ Sequentially “load” part of data; use gradient using “mini-batches” of data
- ▶ Save memory; sometimes even achieve better performance for non-convex problems
- ▶ Uses noisy unbiased gradients

$$x^{(k+1)} = x^{(k)} - \alpha_k \tilde{g}^{(k)}$$

Handwritten notes: A red circle is drawn around $\tilde{g}^{(k)}$ with an arrow pointing to it from the text "estimate" and $\nabla f(x^k)$.

- ▶ $\tilde{g}^{(k)}$ is any noisy unbiased estimate of gradient using “mini-batch” $x^{(k)}$

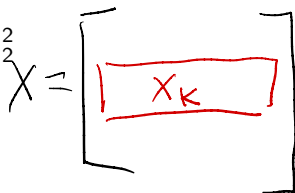
$$\Rightarrow \mathbb{E}[\tilde{g}^{(k)}] = g^{(k)}$$

- ▶ ADAM (2014) is an adaptive learning rate optimization algorithm designed specifically for training deep neural networks.

Stochastic gradient descent for linear regression

- Loss function

$$\min_{\beta} \|y - X\beta\|_2^2$$



A handwritten diagram showing a large square bracket representing a matrix X . Inside this bracket, a smaller rectangle is drawn in red, containing the text X_k . To the left of the large bracket is the letter X with a subscript k .

- Gradient: $f(\beta) = -2X^T(y - X\beta)$
- Draw a random subset of data (X_k, y_k) .
- Stochastic gradient descent:**

$$\rightarrow \beta^{(k+1)} = \beta^{(k)} - 2\alpha_k X_k^T (X_k \beta^{(k)} - y_k)$$

- Compare with Gradient descent:

$$\rightarrow \beta^{(k+1)} = \beta^{(k)} - 2\alpha^{(k)} X^T (X\beta^{(k)} - y)$$

- The approximate gradient is unbiased

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Subgradient

Recall that for convex and differentiable f ,

$$\rightarrow f(y) \geq f(x) + \nabla f(x)^T (y - x) \quad \text{for all } x, y$$

That is, linear approximation always underestimates f .

Definition

A **subgradient** of a convex function f at x is any $g \in \mathbb{R}^n$ such that

$$f(y) \geq f(x) + g^T (y - x) \quad \text{for all } y$$

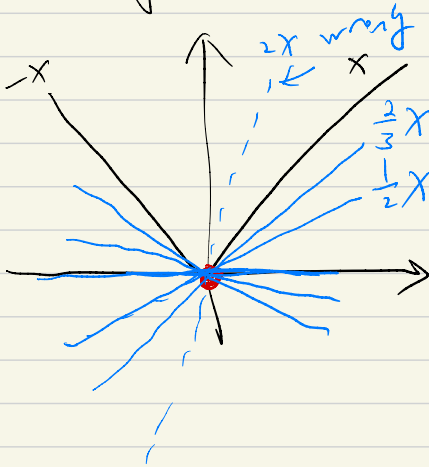
line through $(x, f(x))$ always below function

- Always exists
- If f differentiable at x , then $g = \nabla f(x)$ uniquely
- Same definition works for nonconvex f (however, subgradients need not exist)

Subgradient Examples

- Absolute value: $f(x) = |x|$
- ℓ_2 norm: $f(x) = \|x\|_2$
- ℓ_1 norm: $f(x) = \|x\|_1$
- Maximum of two functions: $f(x) = \max\{f_1(x), f_2(x)\}$ where f_1 and f_2 are convex and differentiable.

$$f(x) = |x|$$



$|x|$ is differentiable when $x \neq 0$

subgradient

$$g = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \\ [-1, 1] & \text{if } x = 0 \end{cases}$$

gradient is a vector

Definition

Set of all subgradients of convex f is called the **subdifferential**:

$$\rightarrow \partial f(x) = \{g \in \mathbb{R}^n : g \text{ is a subgradient of } f \text{ at } x\}$$

- Nonempty (only for convex f)
- $\partial f(x)$ is closed and convex (even for nonconvex f)
- If f is differentiable at x , then $\partial f(x) = \{\nabla f(x)\}$
- If $\partial f(x) = \{g\}$, then f is differentiable at x and $\nabla f(x) = g$

Subgradient Calculus

Basic rules for convex functions:

- Scaling: $\partial(af) = a \cdot \partial f$ if $a > 0$

$$(af)' = a f'$$

- Addition $\partial(f_1 + f_2) = \partial f_1 + \partial f_2$

$$(f_1 + f_2)' = f_1' + f_2'$$

- Affine composition: if $g(x) = f(Ax + b)$, then

$$(f(Ax+b))'$$

$$\partial g(x) = \underbrace{A^T \partial f(Ax + b)} = A \cdot f'(Ax+b)$$

- Finite pointwise maximum: if $f(x) = \max_{i=1,\dots,m} f_i(x)$, then

$$\partial f(x) = \text{conv} \left(\bigcup_{i: f_i(x)=f(x)} \partial f_i(x) \right)$$

convex hull of union of subdifferentials of active functions at x

Subgradient Optimality Condition

$$f \text{ convex} \Rightarrow \exists f(x^*) \leq 0$$

For any f (convex or not),

$$f(x^*) = \min_x f(x) \iff \underline{0 \in \partial f(x^*)}$$

That is, x^* is a minimizer if and only if 0 is a subgradient of f at x^* . This is called the subgradient optimality condition.

Why? Easy: $g = 0$ being a subgradient means that for all y

$$\rightarrow f(y) \geq f(x^*) + 0^T(y - x^*) = f(x^*), \forall y$$

Note the implication for a convex and differentiable function f , with $\partial f(x) = \{\nabla f(x)\}$ KKT Conditions: Min: $\nabla_x L(x, \lambda, \mu) = 0$
 $0 \in \partial_x L(x, \lambda, \mu)$

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Subgradient Method

$$\text{GD: } x^{k+1} \leftarrow x^k - \alpha_k \nabla f(x^k)$$

Now consider f convex, having $\text{dom}(f) = \mathbb{R}^n$, but not necessarily differentiable.

Subgradient method: like gradient descent, but replacing gradients with subgradients. Initialize $x^{(0)}$, repeat:

$$x^{(k+1)} = x^{(k)} - \alpha_k \cdot g^{(k)} \quad k = 1, 2, 3, \dots$$

where $g^{(k)} \in \partial f(x^{(k)})$, any subgradient of f at $x^{(k)}$.

$$d^k = -g^k$$

Subgradient method is not necessarily a descent method, thus we keep track of best iterate $x_{\text{best}}^{(k)}$ among $x^{(1)}, \dots, x^{(k)}$ so far, i.e.,

$$\rightarrow f(x_{\text{best}}^{(k)}) = \min_{i=1, \dots, k} f(x^{(i)})$$

may not
descent
direction

Step Size for Subgradient Method

- Fixed step sizes: $\alpha_k = \bar{\alpha}$ all $k = 1, 2, 3, \dots$
- Diminishing step sizes: choose to meet conditions

$$\sum_{k=1}^{\infty} \alpha_k^2 < \infty, \quad \sum_{k=1}^{\infty} \alpha_k = \infty,$$

Pre determined

i.e., square summable but not summable.

There are several other options too, but key difference to gradient descent: step sizes are pre-specified, **not adaptively computed**.

Convergence Analysis for Subgradient Method

Assume that f is convex, $\text{dom}(f) = \mathbb{R}^n$, and also that f is Lipschitz continuous with constant $L > 0$, i.e., $|f(x) - f(y)| \leq L\|x - y\|_2, \forall x, y$.

Theorem

For a fixed step size t , subgradient method satisfies

$$\lim_{k \rightarrow \infty} f(x_{\text{best}}^{(k)}) \leq f^* + \frac{L^2 \alpha}{2} \quad \text{subconverge}$$

For diminishing step sizes, subgradient method satisfies

$$\lim_{k \rightarrow \infty} f(x_{\text{best}}^{(k)}) = f^*$$

Remark: subgradient method has convergence rate $\mathcal{O}(1/\epsilon^2)$, which is slower than $\mathcal{O}(1/\epsilon)$ rate of gradient descent.